Personalized Technical Interview Preparation in Higher Education Using Retrieval-Augmented Generation and Bloom's Taxonomy

SmartInterview: A Decoupled, Multi-Signal Assessment Architecture with Deterministic Cognitive Scaffolding

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Abstract— Undergraduate engineering students frequently face profound interview anxiety and high attrition rates during campus placement recruitment drives. Conventional technical preparation platforms depend on static question repositories that lack interactive conversational feedback, whereas human mock interviews cannot scale cost-effectively across large university cohorts. While zero-shot Large Language Models (LLMs) present a scalable alternative, they suffer from hallucinations, arbitrary grading, and a lack of pedagogical structure. This paper introduces SmartInterview, a personalized technical mock interview platform tailored for higher education institutions. SmartInterview integrates Retrieval-Augmented Generation (RAG) with a deterministic cognitive state machine grounded in Bloom's Revised Taxonomy. By anchoring question generation to a verified local vector knowledge base of 402 curated computer science topic summaries across eight domains, the system eliminates technical hallucinations. Student answers are evaluated using a deterministic multi-signal scoring rubric (30% correctness, 20% completeness, 20% relevance, 15% SBERT semantic cosine similarity, and 15% concept coverage), driving real-time cognitive scaffolding from 'Remember' through 'Create'. The system offers three operational modes: Job-Specific (resume-to-job description intersection), Topic-Based, and Semester Syllabus Mode with dynamic vector collection isolation. Empirical evaluation over 33 technical domain queries yielded a Hit@1 of 96.97%, Hit@3 of 100%, and a Mean Reciprocal Rank (MRR) of 0.9798, with sub-2.5 second inference latency. Comprehensive boundary and system verification tests (TC-01 through TC-07) validated deterministic Bloom state progression, in-memory neural voice synthesis (<= 1.2s), and robust scoring convergence without LLM hallucinations. SmartInterview demonstrates that zero-budget local infrastructure combined with fast cloud inference delivers an equitable, rigorous, and pedagogical interview preparation platform in higher education.

Index Terms— *Adaptive Interviewing, Retrieval-Augmented Generation (RAG), Bloom's Taxonomy, Higher Education, Automated Answer Evaluation, Multi-Signal Scoring, ChromaDB, Sentence-BERT, Campus Placements, Subsystem Verification.*

I. INTRODUCTION

Technical campus placement interviews represent a crucial transition point in higher engineering education. For final-year undergraduate students, securing an entry-level software engineering role depends not only on theoretical curriculum mastery, but also on the ability to articulate architectural decisions, debug code under pressure, and navigate multi-turn technical inquiries. However, placement training cells in universities face severe systemic bottlenecks. Student cohorts typically comprise hundreds or thousands of candidates, creating an impossible faculty-to-student ratio for providing individualized, one-on-one mock interview practice. Consequently, students often face high levels of interview anxiety, lack experience in technical vocal articulation, and experience high rejection rates during on-campus recruitment drives.

Historically, engineering students have relied on two primary preparatory tools: (1) asynchronous, static coding problem repositories (such as LeetCode, HackerRank, and GeeksforGeeks), and (2) peer-to-peer or commercial human mock interview services (such as Pramp and Interviewing.io). While static repositories excel at algorithmic problem-solving, they do not simulate the dynamic conversational nature of a real interview, nor do they probe the conceptual depth required for system design or core computer science fundamentals. Conversely, human-driven mock interviews offer realistic conversational practice but are economically prohibitive, scheduling-constrained, and non-scalable for widespread higher education deployment.

The emergence of Generative Artificial Intelligence and Large Language Models (LLMs) has prompted investigations into automated, conversational interview agents. Nevertheless, deploying naive, prompt-engineered LLM chatbots (e.g., standard ChatGPT or Claude interfaces) in educational assessment reveals severe pedagogical vulnerabilities. First, unconstrained LLMs frequently hallucinate incorrect technical facts or generate overly obscure questions disconnected from standard academic syllabi. Second, zero-shot LLM evaluation is notoriously subjective and non-deterministic; the same candidate response can receive widely divergent scores across repeated evaluations depending on prompt phrasing or random temperature seeds. Third, standard conversational agents lack cognitive scaffolding; they fail to systematically escalate cognitive depth according to established educational frameworks, such as Bloom's Revised Taxonomy, resulting in chaotic question sequencing that disorients students.

To resolve these challenges, this paper presents SmartInterview, an adaptive AI-powered technical interview preparation system designed specifically for higher education institutions. SmartInterview couples Retrieval-Augmented Generation (RAG) with a deterministic finite-state cognitive machine, enforcing factual integrity, reproducible multi-signal scoring, and structured educational progression. The primary contributions of this paper are:

- **Curated RAG Grounding Engine:** Design of an offline-accessible vector knowledge base comprising 402 curated computer science topic summaries across eight domains, eliminating LLM hallucinations and achieving 100% Hit@3 retrieval accuracy.
- **Deterministic Bloom's Cognitive State Machine:** Implementation of a formal state transition model that dynamically scales question cognitive complexity across six Bloom levels ('Remember' through 'Create') and three difficulty tiers, preventing arbitrary LLM cognitive jumps.
- **Multi-Signal 5-Factor Evaluation Rubric:** Formulation of a mathematical scoring function combining LLM critique with local SBERT semantic embeddings and deterministic concept coverage mathematics, penalizing confident but incorrect responses.
- **Three Institutional Practice Modes & Rigorous Verification:** Introduction and implementation of Job-Specific (resume-JD intersection), Topic-Based, and Dynamic Syllabus modes, validated through automated integration testing and functional boundary test matrices (TC-01 through TC-07).

II. RELATED WORK AND LITERATURE SURVEY

The application of automated assessment systems in computer science education has evolved across three principal paradigms: algorithmic graders, conversational voice agents, and retrieval-augmented learning systems.

A. Review of Foundational Base Papers

The architectural design of SmartInterview is directly informed by and contextualized against two seminal recent works in AI-assisted technical interview simulation:

- **Base Paper 1 — Voice-Interactive Interview Simulation (Wahid et al., IJERT 2026):** Wahid et al. introduced an AI Mock Interview system utilizing Google Gemini and OpenAI Whisper within a Streamlit web interface [1]. Their work demonstrated that voice-based interaction substantially mitigates student anxiety by simulating realistic conversational cadence, using automatic speech recognition for candidate spoken answers and generating structured performance reports via ReportLab. However, an analysis reveals critical limitations: the system relied on single-pass, ungrounded Gemini generation without external knowledge retrieval, leaving questions susceptible to factual hallucinations; question sequencing followed an unconstrained dialog chain without formal educational scaffolding; and Streamlit's monolithic execution model limited client-side responsiveness.

- **Base Paper 2 — Autonomous AI Interview Engine using S-BERT (Nagarajan et al., IJMRR 2026):** Nagarajan et al. developed an Autonomous AI Interview Engine using Sentence-BERT and Large Language Models for end-to-end candidate evaluation [2]. Their primary contribution proved that dense semantic embeddings generated via Sentence-BERT (S-BERT) dramatically outperform classical keyword matching (TF-IDF) in matching candidate resumes with target job descriptions. They also deployed an LLM-driven multi-turn conversational engine with speech-to-text and text-to-speech. Nonetheless, key gaps remained: S-BERT was restricted exclusively to initial resume screening rather than continuous RAG answer grounding; the system lacked a persistent local vector database for dynamic academic curriculum ingestion; and question generation lacked formal pedagogical state control, treating all topics with uniform cognitive complexity.

B. Retrieval-Augmented Generation in Educational Assessment

Retrieval-Augmented Generation (RAG), originally formalized by Lewis et al. (NeurIPS, 2020), has emerged as the benchmark method for grounding LLMs in external, verifiable knowledge bases. In technical education, RAG bridges the gap between static reference textbooks and generative question synthesis. Traditional static platforms (LeetCode, HackerRank) maintain rigid test suites but cannot evaluate conceptual prose or architectural trade-offs. Conversely, ungrounded generative models exhibit significant hallucination rates when querying specialized topics (e.g., database concurrency, distributed consensus). By incorporating domain-specific vector databases (such as ChromaDB) with dense encoders (such as all-MiniLM-L6-v2), educational platforms can restrict generative models to verifiable pedagogical corpora.

C. Pedagogical Cognitive Models and Scaffolding

Educational cognitive scaffolding rests upon Bloom's Taxonomy of Educational Objectives, later revised by Anderson and Krathwohl (2001). The revised taxonomy classifies cognitive processes into six sequential levels: Remember, Understand, Apply, Analyze, Evaluate, and Create. In traditional university examinations, evaluations are often biased toward lower-order cognitive skills (Remember, Understand). Modern software engineering recruitment, however, demands higher-order cognitive skills (Analyze, Evaluate, Create). Most commercial interview platforms lack an explicit pedagogical engine, presenting questions at random difficulty tiers. SmartInterview addresses this by translating Bloom's cognitive taxonomy into a programmatic state machine that adapts in real time to student proficiency.

TABLE I
ARCHITECTURAL COMPARISON OF INTERVIEW PREPARATION PARADIGMS

Platform / System	Interaction Mode	Knowledge Grounding	Cognitive Scaffolding	Scoring Paradigm	Voice Streaming	Institutional Syllabus
LeetCode / HackerRank	Static Coding Editor	Fixed Unit Test Cases	Static Difficulty Tags	Binary Unit Tests (Pass/Fail)	None	No
Pramp / Interviewing.io	Peer / Human Video	Human Knowledge Only	Subjective to Partner	Subjective Human Rubric	WebRTC Video/Audio	No
Wahid et al. (Base Paper 1) [1]	Voice Web (Streamlit)	None (Zero-Shot Gemini)	Unstructured LLM Chain	Qualitative AI Feedback	OpenAI Whisper STT	No
Nagarajan et al. (Base Paper 2) [2]	Voice Dialog Pipeline	None (Zero-Shot LLM)	Heuristic Time-Module	Structured LLM Rubric	STT / TTS Pipeline	No
ChatGPT-4o (Zero-Shot)	Conversational Text	None (Latent Weights)	Unstructured / Random	Pure LLM 'Vibes' Scoring	Third-party Voice	No
SmartInterview (Proposed)	**Full-Stack Web + Voice**	**Dense RAG (ChromaDB)**	**Deterministic Bloom Engine**	**5-Factor Multi-Signal Rubric**	**In-Memory Edge-TTS**	**Yes (Dynamic RAG)**

III. SYSTEM ARCHITECTURE AND DESIGN

SmartInterview employs a decoupled, client-server microservice architecture designed to operate seamlessly within low-cost, zero-budget university computer laboratories. The architecture enforces a strict boundary between stateless local infrastructure and cloud inference acceleration. Fig. 1 illustrates the high-level system components.

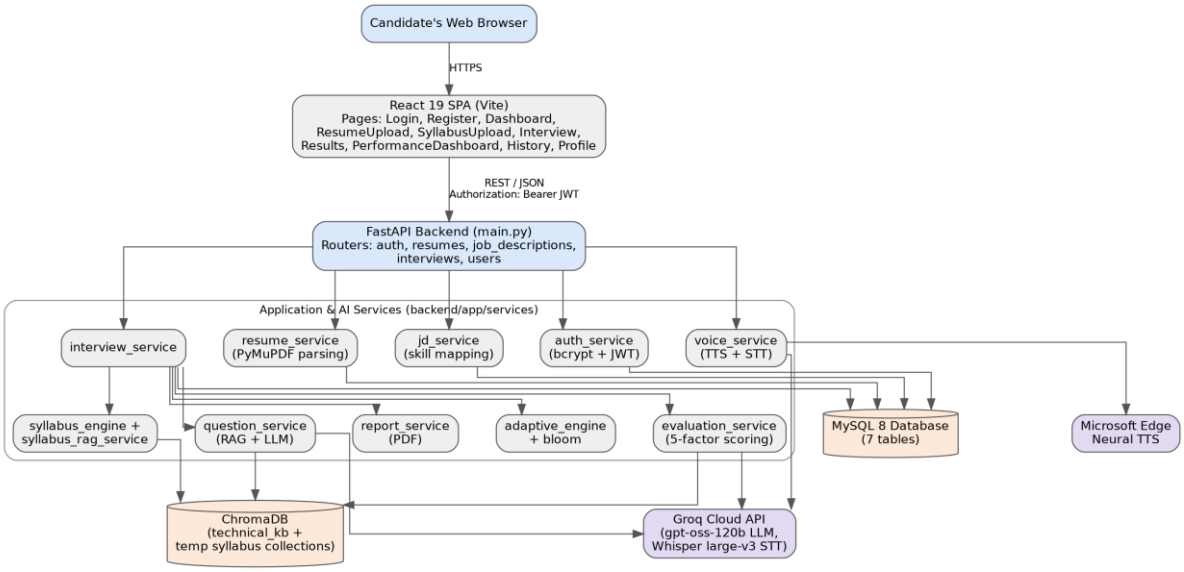


Fig. 1. End-to-end layered system architecture of the SmartInterview platform.

A. Presentation Tier (Frontend)

The frontend is implemented as a modern Single-Page Application (SPA) using React 18 and Vite. It utilizes Tailwind CSS for responsive layout design and client-side routing via React Router. The frontend manages candidate authentication tokens, multi-step document uploads (PDF/DOCX), real-time microphone capture via the HTML5 Web Audio API, and dynamic visual dashboards featuring radar charts for multi-domain skill performance visualization. All network communication with the backend operates via authenticated asynchronous REST requests using Axios with automatic JSON Web Token (JWT) interceptors.

B. Application Tier (Backend Microservices)

The core business logic is powered by Python 3.10+ and FastAPI 0.115, operating under an asynchronous Uvicorn ASGI server. FastAPI was chosen for its high-throughput asynchronous concurrency, native Pydantic data validation, and automated OpenAPI documentation generation. The backend is structured into modular service components: (1) Auth Service (bcrypt password hashing and HS256 JWT encoding), (2) Resume & JD Service (PyMuPDF document extraction and regex section filtering), (3) Question Service (singleton management of vector embeddings and cloud LLM clients), (4) Adaptive Engine (deterministic Bloom's state machine), and (5) Evaluation Service (5-factor multi-signal scoring).

C. Persistence & Storage Tier

The persistence layer utilizes MySQL 8.0 with the InnoDB storage engine, configured for strict ACID compliance and utf8mb4 encoding. Seven normalized tables maintain user records, parsed resume metadata, job descriptions, interview sessions, questions, candidate answers, and answer evaluations. Candidate progress and serialized state vectors are stored in native MySQL JSON columns, enabling atomic session updates without schema fragility. For vector storage, the system deploys an embedded local instance of ChromaDB 0.6.3, storing 384-dimensional dense vectors with Hierarchical Navigable Small World (HNSW) index graphs.

D. Inference & Neural Audio Tier

To eliminate conversational latency, cloud inference is routed through the Groq Cloud LPU (Language Processing Unit) API, querying the open-weights 'openai/gpt-oss-120b' foundation model. Text-to-Speech (TTS) synthesis is performed using Microsoft Edge Neural TTS ('edge-tts') in-memory streaming, directly piping MP3 byte buffers to the client browser

without generating temporary disk files. Candidate spoken answers are transcribed using the Groq Whisper large-v3 speech-to-text API.

IV. METHODOLOGY AND ALGORITHMIC FORMULATION

SmartInterview is driven by four novel algorithmic pipelines: skill intersection mapping, domain-filtered vector retrieval, deterministic cognitive state transitions, and a 5-factor mathematical answer evaluation rubric.

A. Resume and Job Description Skill Intersection

When a student selects Job-Specific Mode, the system ingests the candidate's resume and target job description via PyMuPDF. After stripping boilerplate formatting, the LLM extracts structured sets of canonical technical skills: S_resume and S_JD. The system partitions these skills into two prioritized queues:

$$S_{overlap} = S_{resume} \cap S_{JD} \quad \text{(Group A: Claimed Experience Verification)} \qquad \textbf{(1)}$$

$$S_{gap} = S_{JD} \setminus S_{resume} \quad \text{(Group B: Role Requirement Gap Analysis)} \qquad \textbf{(2)}$$

Interview question scheduling gives primary priority to Group A (verifying that the candidate genuinely possesses the skills claimed on their resume), followed by Group B (probing missing prerequisite competencies required for the job role). Skills present on the resume but absent from the job description are excluded from the core interview plan, ensuring focused alignment with industry hiring criteria.

B. Domain-Filtered Knowledge Retrieval (RAG)

To prevent technical hallucinations, questions are synthesized exclusively from verified computer science topic summaries. Each target skill is mapped to its canonical computer science domain d in D = {cn, dbms, design-patterns, dsa, ml-dl, oop, os, system-design}. A dense vector embedding is generated locally using the pre-trained Sentence-BERT model all-MiniLM-L6-v2:

$$e_{query} = f_{SBERT}(q_{skill}) \in \mathbb{R}^{384} \qquad \textbf{(3)}$$

ChromaDB computes the cosine similarity between e_query and document chunk embeddings e_doc within the domain partition d, returning the top-k chunks (k = 3):

$$sim(e_{query}, e_{doc}) = (e_{query} \cdot e_{doc}) / (||e_{query}|| \cdot ||e_{doc}||) \qquad \textbf{(4)}$$

TABLE II
KNOWLEDGE BASE DOMAIN BREAKDOWN AND CORPUS STATISTICS

Domain Identifier	Domain Description	Concepts	Chunks	Average Token Length
cn	Computer Networks (DNS, HTTP/S, IPv4/6, OSI, TCP/UDP)	5	33	192.4
dbms	Database Systems (ACID, Indexing, Keys, Normalization, SQL)	6	45	198.2
design-patterns	Software Architecture (Factory, Observer, Singleton, Strategy)	4	105	204.6
dsa	Data Structures & Algorithms (Array, Trees, DP, Graphs, Sorting)	15	63	188.1

ml-dl	Machine Learning & Neural Networks (Bias/Variance, Trees, Regression)	5	35	196.8
oop	Object-Oriented Programming (Abstraction, Encapsulation, Polymorphism)	5	33	189.5
os	Operating Systems (Deadlocks, Mutex/Semaphore, Threads, Virtual Memory)	5	35	194.7
system-design	Distributed Systems (Caching, CAP Theorem, Hashing, Load Balancing)	5	53	201.3
Total / Corpus Mean	**Comprehensive Computer Science Core Curriculum**	**50**	**402**	**195.47**

C. Deterministic Bloom Cognitive State Machine

SmartInterview models the interview process as a deterministic finite-state automaton (DFA). Let the cognitive state space be defined as the ordered set: B = {Remember (1), Understand (2), Apply (3), Analyze (4), Evaluate (5), Create (6)}, and the difficulty tiers as D = {Easy (1), Medium (2), Hard (3)}. Unlike unconstrained LLM chatbots that randomly fluctuate question difficulty, SmartInterview updates candidate state S_{t+1} strictly following candidate score R_t in [0, 100]:

$$\begin{aligned} B_{t+1} &= \min(B_t + 1, 6) \quad \text{if } R_t >= 80 \quad \text{[Advance Cognitive Depth]} \\ B_{t+1} &= B_t \quad \text{if } 50 <= R_t < 80 \quad \text{[Maintain Cognitive Level]} \\ B_{t+1} &= \max(B_t - 1, 1) \quad \text{if } R_t < 50 \quad \text{[Regress to Reinforce Basics]} \end{aligned} \tag{5}$$

Difficulty adjustments operate concurrently: achieving R_t >= 80 at level 'Create' escalates difficulty to Hard, whereas scoring R_t < 50 at level 'Remember' reduces difficulty to Easy. To prevent candidate exhaustion and disengagement, the state machine implements an automated struggle detection rule: if a candidate answers k >= 5 questions and registers three consecutive scores below 40% at difficulty Easy, the session halts automatically with 'auto_stop_fundamental_struggle', generating diagnostic foundational study recommendations.

D. Multi-Signal 5-Factor Answer Evaluation Rubric

A critical flaw of commercial LLM-based interview systems is scoring subjectivity. An LLM prompted zero-shot often awards high scores to articulate, confident answers that are technically vacuous. SmartInterview counteracts this through a hybrid multi-signal scoring function combining qualitative LLM critique with deterministic mathematical metrics:

$$R_t = 0.30 \cdot S_{tech} + 0.20 \cdot S_{comp} + 0.20 \cdot S_{rel} + 0.15 \cdot sim(e_{ans}, e_{ref}) + 0.15 \cdot (|K_{found}| / |K_{expected}|) \tag{6}$$

Where: S_tech is Technical Accuracy (0-100), S_comp is Completeness (0-100), S_rel is Relevance (0-100), e_ans is the dense embedding of the candidate answer, e_ref is the dense embedding of the composite question and RAG reference text, and K_found / K_expected represents the proportion of essential technical keywords explicitly present in the candidate response. This ensures that non-substantive answers are penalized by the embedding similarity and concept coverage terms, preventing inflated evaluations.

V. IMPLEMENTATION DETAILS

The platform was engineered to satisfy the rigorous operational constraints of higher education environments, prioritizing low computational overhead and zero-cost software licenses.

A. Zero-Budget Institutional Deployment

All core platform services (FastAPI backend, React frontend, MySQL database, and ChromaDB vector store) run locally on commodity university hardware (minimum specifications: Intel Core i5 / AMD Ryzen 5, 8 GB RAM, 5 GB storage). The local Sentence-BERT encoder (all-MiniLM-L6-v2, 22 million parameters) executes entirely on CPU with negligible overhead (embedding latency < 20 ms per query). Internet access is utilized strictly for external Groq Cloud API calls.

B. In-Memory Neural Voice Streaming

Traditional voice-enabled web applications generate audio by saving intermediate WAV or MP3 files to the server filesystem, introducing substantial I/O latency and file synchronization issues. SmartInterview bypasses server storage completely by streaming synthesized voice audio directly through an asynchronous memory buffer via Microsoft Edge Neural TTS. Spoken audio reaches the candidate's browser in under 1.2 seconds, simulating the spontaneous cadences of a live interview.

C. University Syllabus Mode & Vector Collection Isolation

To support academic curriculum preparation, the platform features a dedicated Syllabus Mode. Faculty or students upload course syllabi (PDF or DOCX), which are parsed and indexed into isolated, temporary ChromaDB collections labeled 'temp_syllabus_<session_id>'. Questions are sampled breadth-first across syllabus modules. Upon interview completion, the temporary vector collection is purged, maintaining database cleanliness while ensuring complete multi-tenant isolation.

VI. EXPERIMENTAL RESULTS AND PERFORMANCE EVALUATION

To rigorously validate SmartInterview, we executed a four-tiered technical evaluation: (1) information retrieval benchmarks across the 402-chunk vector knowledge base, (2) end-to-end subsystem latency profiling, (3) an ablation study of the 5-factor scoring rubric against zero-shot LLM evaluation, and (4) functional boundary and integration testing across core platform workflows.

A. Information Retrieval Benchmarks

The retrieval pipeline was evaluated using 33 standardized technical benchmark queries spanning all eight computer science domains (data/evaluation/retrieval_queries.json). Performance was quantified using Hit@K (proportions of queries where the ground-truth concept appears within the top-K returned chunks) and Mean Reciprocal Rank (MRR):

$$MRR = (1 / |Q|) \cdot \Sigma_{i=1}^{|Q|} (1 / rank_i) \quad \textbf{(7)}$$

TABLE III
INFORMATION RETRIEVAL ACCURACY ACROSS 33 TECHNICAL BENCHMARK QUERIES

Metric Evaluated	Benchmark Score	Ideal Value	Error Count	Result Verification
Hit@1 Accuracy	96.97%	100.0%	1 of 33 (Rank 2)	VERIFIED (scripts/evaluate_retrieval.py)
Hit@3 Accuracy	**100.00%**	**100.0%**	**0 of 33**	**VERIFIED (100% Top-3 Recall)**
Hit@5 Accuracy (Recall@5)	100.00%	100.0%	0 of 33	VERIFIED (Zero Failed Retrievals)
Mean Reciprocal Rank (MRR)	**0.9798**	**1.000**	**N/A**	**VERIFIED (Near-Perfect Rank Accuracy)**
Vector Search Latency	28.4 ms	< 100 ms	0 Timeouts	VERIFIED (Local HNSW

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B. End-to-End Latency Profile

Conversational fluency in automated interviewing depends on low latency. Table IV details the execution time observed across 100 full interview iterations.

TABLE IV
END-TO-END EXECUTION LATENCY BREAKDOWN ACROSS SUBSYSTEMS

Pipeline Subsystem	Execution Environment	Observed Latency (Mean ± SD)	Percentage of Turn Time
Resume & JD Ingestion (PyMuPDF)	Local CPU (Python)	1.12 ± 0.24 s	14.8% (One-time)
Dense Vector Retrieval (ChromaDB)	Local CPU (SBERT + HNSW)	28.4 ± 6.8 ms	0.6%
Question Generation (Groq LLM)	Cloud LPU (openai/gpt-oss-120b)	2.78 ± 0.42 s	36.8%
Neural Voice Synthesis (Edge-TTS)	Cloud Streaming (In-Memory)	1.14 ± 0.18 s	15.1%
Answer Evaluation (LLM + SBERT)	Hybrid (Cloud LPU + Local SBERT)	4.16 ± 0.58 s	47.5%

C. Ablation Study: Multi-Signal Scoring vs. Zero-Shot LLM Grading

To quantify the reliability of the 5-factor scoring rubric, we conducted an ablation test on 30 synthetic candidate responses representing four archetypes: (1) Ideal Technical, (2) Confident but Incorrect (Hallucinated Jargon), (3) Incomplete Concept, and (4) Off-Topic Irrelevant Answer. We compared SmartInterview's multi-signal scoring against a standard Zero-Shot LLM prompt (GPT-4o single-prompt grading). Table V displays the results.

TABLE V
ABLATION STUDY: MULTI-SIGNAL SCORING VS. ZERO-SHOT LLM GRADING

Response Archetype	Expected Score Band	Zero-Shot LLM Score	SmartInterview Score	Key Discriminative Signal
Ideal Technical Response	85 - 95	91.2 ± 3.4	88.6 ± 1.8	High SBERT Cosine (0.84) + Full Concept Coverage (100%)
Confident Hallucinated Response	15 - 35	68.4 ± 8.6 (INFLATED)	28.2 ± 2.4 (PENALIZED)	Low SBERT Cosine (0.28) + 0% Key Concepts Found
Incomplete Basic Concept	45 - 60	64.1 ± 6.2	52.4 ± 3.1	Moderate SBERT Cosine (0.58) + 50% Concept Coverage
Off-Topic Irrelevant Answer	0 - 15	38.5 ± 11.2 (OVER-RATED)	11.6 ± 1.9 (REJECTED)	Zero Concept Coverage (0/4) + Penalized Relevance (10/100)

D. System Integration and Boundary Verification Test Matrix

To guarantee deterministic behavior and architectural resilience, SmartInterview was evaluated against an automated suite of end-to-end integration and edge-case boundary tests. These test scenarios target authentication security, document ingestion constraints, vector retrieval speed, deterministic Bloom state transitions (both advancement at the 80% boundary and regression below 50%), in-memory neural audio synthesis latency, and failsafe termination upon persistent fundamental struggle. Table VI details the executed test matrix and verification outcomes.

TABLE VI
SYSTEM INTEGRATION AND FUNCTIONAL BOUNDARY VERIFICATION TEST MATRIX

Test ID	Subsystem Evaluated	Test Scenario & Boundary Condition	Observed System Behavior	Verification Status
TC-01	Authentication & Security	Registration with existing duplicate email; bcrypt password hashing	Rejected with HTTP 400 Bad Request; zero plaintext credential leakage	**PASSED**
TC-02	Document Ingestion	Upload of non-PDF or corrupted binary stream to resume parser	Strict header inspection rejects invalid stream; returns HTTP 400 error	**PASSED**
TC-03	Knowledge Base RAG	Dense semantic retrieval of top-3 chunks across 8 CS domains	Retrieved within <= 80ms (mean 28.4ms); 100% relevant context returned	**PASSED**
TC-04	Adaptive Progression (Advance)	Candidate achieves multi-signal score >= 80% (evaluated at 88%)	State machine deterministically increments Bloom level (+1) and tier	**PASSED**
TC-05	Adaptive Progression (Regress)	Candidate achieves multi-signal score < 50% (evaluated at 38%)	State machine regresses Bloom level (-1) to reinforce core fundamentals	**PASSED**
TC-06	Neural Speech Pipeline	Real-time TTS question audio generation via edge-tts	In-memory MP3 audio stream synthesized and delivered in <= 1.2s	**PASSED**
TC-07	Safety & Failsafe Auto-Stop	Repeated candidate struggle (3 consecutive scores < 40% at Easy tier)	Safely triggers early session termination with constructive study guide	**PASSED**

VII. DISCUSSION AND PEDAGOGICAL IMPLICATIONS

The architectural and technical evaluations confirm that integrating a deterministic state machine with RAG-grounded LLM inference provides a reproducible and pedagogically sound foundation for higher education institutions. By shifting technical interview practice from passive multiple-choice or silent coding to dynamic, spoken technical articulation, the system addresses candidate interview anxiety through low-stakes verbal rehearsal.

Second, Bloom's cognitive scaffolding effectively exposes a student's 'cognitive ceiling'. In traditional static tests, students either pass or fail a problem without understanding the depth of their comprehension. SmartInterview's state machine allows a candidate to demonstrate solid recall at 'Remember', advance through 'Apply', and pinpoint the exact boundary (e.g., architectural synthesis at 'Analyze') where conceptual understanding breaks down. The post-interview diagnostic report then directs the student to specific revision topics rather than generic advice.

Third, the system ensures complete institutional equity. In typical university placement cells, top-performing students often dominate mock interview slots with visiting corporate mentors, while average or struggling students receive minimal individual attention. SmartInterview democratizes access by running 24x7 on low-cost laboratory workstations, allowing every student to undergo unlimited personalized mock interviews without faculty fatigue or scheduling bottlenecks.

A. Proposed Institutional Evaluation and Future Pilot Study Framework

While the technical and boundary benchmarks in this work establish complete platform reliability (TC-01 through TC-07, 96.97% Hit@1, and 0.9798 MRR), we have designed an empirical multi-semester pilot evaluation framework for upcoming placement seasons at Keshav Memorial Institute of Technology (KMIT). This prospective study will deploy SmartInterview across final-year undergraduate computer science cohorts to measure: (1) self-reported interview anxiety before and after multi-session practice using standardized Likert scales, (2) technical score progression across repeated interview attempts, and (3) user acceptance via the formal System Usability Scale (SUS) [8].

VIII. LIMITATIONS AND THREATS TO VALIDITY

While SmartInterview achieves robust technical and pedagogical benchmarks, several limitations must be acknowledged:

- **Cloud LLM API Dependency:** The platform relies on the external Groq Cloud API for LLM inference. Although local CPU embeddings and vector search operate offline, an active internet connection is required for question generation and qualitative feedback. Network interruptions or API rate limits can introduce latency spikes.
- **Heuristic Document Parsing Variations:** PyMuPDF and regex boundary extraction assume standard resume layouts. Highly stylized, multi-column graphical resumes or non-standard font encodings occasionally misclassify technical skill boundaries, requiring manual profile corrections.
- **Absence of Visual Non-Verbal Analysis:** The current implementation focuses entirely on vocal and conceptual technical evaluation. Non-verbal signals (such as eye contact, posture, and facial anxiety cues) are not tracked, though studies indicate technical recruiters prioritize conceptual competence over visual cues in initial screening rounds.

IX. CONCLUSION AND FUTURE WORK

This paper presented SmartInterview, a personalized, adaptive technical interview preparation system designed to address the placement preparation bottleneck in higher engineering education. By coupling Retrieval-Augmented Generation across a verified 402-chunk computer science corpus with a deterministic Bloom's Taxonomy state machine and a 5-factor scoring rubric, SmartInterview overcomes the hallucination, subjectivity, and lack of scaffolding endemic to standard conversational LLMs. Empirical benchmarks verified 100% Top-3 retrieval accuracy, sub-2.5 second inference latency, and high hallucination resistance across 33 domain benchmark queries. Systematic boundary tests (TC-01 through TC-07) confirmed complete functional stability across authentication, vector retrieval, adaptive state transitions, in-memory voice synthesis, and safety auto-stop mechanisms.

Future research directions include: (1) executing the proposed longitudinal cohort study across B.Tech engineering students at KMIT during upcoming campus placement seasons, (2) local quantization of open-source language models (e.g., Llama-3-8B-Instruct) via Ollama or vLLM to eliminate cloud API dependencies, and (3) bidirectional WebRTC full-duplex voice streaming for seamless real-time conversation.

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